Horizon-Maintenance Dark Energy: A Pre-Euclid Bridge Note from Finite Distinguishability

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(S-X1 / Conditional Physical Bridge / Pre-Euclid Version)

This note registers a frontier physical consequence (P3 layer) connecting finite distinguishability at cosmological horizons to dark-energy phenomenology before Euclid Data Release 1. The claim is not that the cosmological constant, general relativity, quantum gravity, or the dark-energy equation of state is derived from the distinction primitive alone. Rather, if a cosmological horizon is treated as a finite distinguishability boundary with horizon entropy and temperature, then the natural energy-density scale of horizon maintenance is

$$\rho_{\mathrm{HM}} \sim \frac{c^4}{GR_{\mathrm{h}}^2} \sim \frac{c^2H^2}{G} \qquad (R_{\mathrm{h}} \sim c/H),$$

or, in natural units, $\rho_{\mathrm{HM}} \sim M_{\mathrm{Pl}}^2 H^2$. This gives the observed smallness scale $\rho_{\mathrm{HM}}/\rho_{\mathrm{Pl}} \sim (\ell_{\mathrm{P}}/R_{\mathrm{h}})^2 \sim 10^{-122}$, not by cancellation of large vacuum contributions but by boundary-area suppression. The note formulates a conservative phenomenological model $\rho_{\mathrm{HM}}(a) = 3M_{\mathrm{Pl,red}}^2 H^2(a)\mu(a)$, where $\mu(a)$ is a horizon-maintenance response factor, and pre-registers a falsifiable expectation: the physical horizon-maintenance component should be dynamical but non-phantom, $w_{\mathrm{phys}} \geq -1$, with late-time approach toward -1. Apparent phantom behavior in effective CPL or other reconstructions is allowed only as an effective description arising from sector mixing, parameterization, systematics, curvature, modified-gravity mapping, or an incomplete correspondence between fitted and physical dark-energy variables. The bridge is demoted if Euclid, DESI, Roman, CMB-S4, or combined nonparametric reconstructions lock dark energy to a strict constant with high precision. The strict non-phantom version is falsified by robust, unavoidable physical $w < -1$ behavior or by a dark-energy density that unambiguously increases with scale factor under a model-independent reconstruction. The purpose is pre-registration, auditability, and model development, not replacement of ΛCDM.

Reader Contract. This paper is a frontier physical consequence note (P3), not a completed cosmological theory. It does not claim to derive general relativity, quantum gravity, the Bekenstein-Hawking entropy formula, the exact numerical value of the cosmological constant, or a complete likelihood-ready dark-energy model. It uses established horizon entropy, horizon temperature, finite causal access, and Landauer-style update accounting as bridge inputs. Its contribution is to register a finite-distinguishability interpretation of the dark-energy scale, state conservative predictions, and specify advance falsification and demotion conditions before Euclid DR1.

Claim-status summary

Keywords: dark energy; finite distinguishability; cosmological horizon; horizon entropy; Landauer principle; holographic dark energy; equation of state; Euclid; DESI; pre-registration.

INTRODUCTION

Pre-Euclid purpose of this note

Euclid Data Release 1 is currently scheduled for October 21, 2026, after earlier quick releases intended to prepare the community for larger validated data products [23, 24]. DESI has already reported strengthened hints that dark energy may evolve, based on its first three years of BAO data and combinations with other cosmological probes [21, 22]. These results motivate theoretical discipline. If a theory has a dark-energy claim, it should state its prediction and failure conditions before the next major data release rather than reinterpret outcomes afterward.

This paper therefore registers FDS-X1 as a *pre-Euclid frontier physical consequence (P3)*. It is not a completed replacement for ΛCDM. It is a timestamped claim package: a scale argument, a minimal phenomenological model, a set of pre-registered expectations, and a falsification table.

TABLE I. Summary of central FDS-X1 claims, their epistemic status, and what would constitute falsification or demotion.

Claim	Status	What would fail
Cosmological horizon as finite distinguishability boundary	Physical bridge	Horizon entropy/causal-access accounting fails or is superseded in the applicable regime
Horizon-maintenance density scale $\rho_{\mathrm{HM}} \sim c^4/(GR_h^2)$	Scaling bridge	Dark-energy scale is shown to be unrelated to any horizon-area or causal-boundary scale
10^{-122} as area-ratio suppression	Naturalness argument	Observed scale requires independent cancellation rather than boundary-area suppression
Dynamical but non-phantom physical component	Pre-Euclid prediction	Robust unavoidable physical $w < -1$, or robust $d\rho_{\mathrm{DE}}/d\ln a > 0$, not attributable to effective reconstruction
Late-time approach toward -1	Asymptotic bridge	Future data favor persistent non-de Sitter behavior incompatible with horizon stabilization
Strict Λ demotion criterion	Failure contract	High-precision, model-independent consistency with constant ρ_{DE} over the relevant redshift range

Central idea

In FDS-T1, a finite physical observer has an observer-relative distinguishability budget $N_O = |\operatorname{Im}(\pi_O)|$ and an accessible bit capacity limited by memory, boundary, channel, causal access, and update throughput [1]. A cosmological horizon is a limiting physical case of such finite access. It separates what can in principle affect an observer's records from what cannot.

The FDS-X1 bridge claim is that the dark-energy scale can be interpreted as a horizon-maintenance scale: the energy-density scale associated with sustaining a finite causal distinguishability boundary. If the relevant horizon has radius R_h, area A_h, entropy $S_h \sim k_B A_h/(4\ell_P^2)$, and temperature $T_h \sim \hbar c/(2\pi k_B R_h)$, then the natural boundary-maintenance energy density scales as

$$\rho_{\mathrm{HM}} \sim \frac{c^4}{GR_h^2}. \tag{1}$$

For $R_h \sim c/H$, this becomes

$$\rho_{\mathrm{HM}} \sim \frac{c^2 H^2}{G}, \tag{2}$$

or $\rho_{\mathrm{HM}} \sim M_{\mathrm{Pl}}^2 H^2$ in natural units. This is the same scale as the critical density up to order-unity factors. The observed dark-energy density is therefore not treated as a Planck-volume vacuum sum; it is treated as a horizon-scale boundary term.

What is not claimed

This note does not claim any of the following:

1. that FDS derives general relativity or quantum gravity;
2. that FDS derives the coefficient 1/4 in the Bekenstein-Hawking entropy formula;
3. that the cosmological constant problem is solved in the sense of a complete microphysical vacuum-energy cancellation;
4. that the exact value of w_0, w_a is predicted without a full likelihood-ready cosmological model;
5. that DESI has already confirmed FDS-X1;
6. that any deviation from ΛCDMcounts as support for FDS-X1;
7. that apparent phantom reconstructions are automatically artifacts.

The note is narrower: it registers a scaling bridge, a physical interpretation, a minimal model family, and advance demotion rules.

Relation to FDS-T1 and the Physical Bridge Registry

FDS-X1 depends on the finite-observer budget language of FDS-T1 [1], the FDS formal core [2], the DT archive [3], and the Physical Bridge Claim Registry [4]. FDS-T1 supplies the shared concepts of finite distinguishability, boundary-relative capacity, and Landauer-style update accounting. The Physical Bridge Registry classifies dark-energy claims as frontier physical consequences (P3 layer) rather than established consequences of the formal core. FDS-X1 therefore cannot be used as evidence for the FDS core unless it later survives independent cosmological tests; failure of X1 would demote or retire the dark-energy bridge, not the finite-system formalism.

RELATED WORK

The cosmological constant problem

The accelerated expansion of the universe was inferred from Type Ia supernova observations [5, 6] and is supported by CMB and large-scale-structure measurements [7]. In standard cosmology, the simplest explanation is a cosmological constant Λ, with dark-energy equation of state $w = -1$. The cosmological constant problem arises because naive quantum-field vacuum-energy estimates are enormously larger than the observed dark-energy density. The present paper does not compute or cancel zero-point energies. It asks whether the observed scale is more naturally associated with a horizon boundary than with independent bulk vacuum modes.

Horizon thermodynamics

Black-hole thermodynamics assigns entropy and temperature to horizons [8, 9]. Gibbons and Hawking showed that de Sitter horizons also carry a temperature and entropy [10]. Jacobson's thermodynamic derivation of Einstein's equation suggests a deep relation between local horizon thermodynamics and gravitational dynamics [15]. FDS-X1 does not derive these results. It uses them as physical bridge inputs for a finite-distinguishability interpretation of horizon-scale energy density.

Holographic and horizon dark energy

The idea that dark-energy scales may be controlled by infrared or horizon cutoffs is not new. Cohen, Kaplan, and Nelson argued that effective field theories in a box must avoid states whose Schwarzschild radius exceeds the box size, leading to UV/IR relations [17]. Hsu pointed out problems with taking the Hubble radius as the infrared cutoff in simple holographic dark-energy models [18]. Li proposed holographic dark energy using the future event horizon as the infrared cutoff, producing late-time acceleration in a phenomenological model [19]. Reviews summarize the broad holographic dark-energy program and its observational constraints [20].

FDS-X1 is adjacent to this literature but not identical. It does not merely impose an IR cutoff on quantum fields. It interprets the horizon as a finite distinguishability boundary whose maintenance scale is boundary-like rather than volume-like. This distinction matters because the FDS claim concerns the accessible distinction budget of a finite observer patch, not a bare count of field modes.

Novelty boundary relative to holographic dark energy

FDS-X1 does not claim priority over the holographic dark-energy scaling relation $\rho_{\mathrm{DE}} \sim M_{\mathrm{Pl}}^2 L^{-2}$. That scaling and the non-phantom tendency $w > -1$ are well established in the holographic dark-energy literature [17–20]. The novelty of the FDS-X1 bridge lies in the finite-distinguishability interpretation of that scale and its embedding in a broader FDS boundary-accounting framework.

Current observational context

DESI's 2025 three-year results strengthened hints that dark energy may be evolving [21]. The DESI DR2 guide states that the evidence for a time-evolving dark-energy equation of state increased relative to Year 1 and that extended analyses indicate trends across parametric and non-parametric methods [22]. At the same time, such hints remain data-combination and model dependent, and the community has not replaced ΛCDM. Euclid DR1, planned for October 21, 2026, is therefore a natural target for a pre-registered status update [23, 24].

FINITE-DISTINGUISHABILITY SETUP

Horizon as an access boundary

Definition 1 (Cosmological distinguishability boundary). *A cosmological distinguishability boundary for an observer O is a horizon-like surface beyond which physical differences cannot be operationally registered, preserved, and used by O within the relevant causal and thermodynamic constraints.*

The relevant boundary need not be exactly the Hubble radius. Depending on the problem, it may be an apparent horizon, event horizon, causal diamond, or other effective boundary. In a spatially flat FRW universe, common length scales include the Hubble radius $R_H = c/H$, the apparent horizon $R_A = c/H$, and, in accelerating spacetimes, the future event horizon

$$R_e(t) = a(t) \int_t^\infty \frac{c\, dt'}{a(t')}. \tag{3}$$

In exact de Sitter space, these coincide up to the common value c/H. During matter domination or transition eras, they need not coincide. FDS-X1 therefore uses R_h as a generic effective horizon radius and treats the mapping $R_h \mapsto H^{-1}$ as a late-time or phenomenological approximation, not an identity valid in every epoch.

TABLE II. FDS-X1 novelty boundary relative to holographic dark energy.

Aspect	HDE prior art	FDS-X1 addition
$\rho \sim M_{\rm Pl}^2 L^{-2}$	established	not claimed as novel
horizon / IR cutoff	established	interpreted as finite distinguishability boundary
non-phantom $w > -1$ possibility	established	registered as physical horizon-maintenance constraint
10^{-122} area-ratio smallness	present in HDE background	interpreted as boundary-area suppression of maintainable distinctions
cutoff choice	model-dependent selection	constrained by causal-access / recordability / boundary-maintenance cr
failure conditions	not always explicit	pre-registered demotion / falsification table
connection to geometry bridge	partial in emergent gravity	T2 supplies effective geometry as horizon ledger

Horizon entropy and temperature

Assume a horizon-like boundary with area

$$A_h = 4\pi R_h^2. \tag{4}$$

The associated entropy is

$$S_h = \frac{k_B A_h}{4\ell_P^2}, \tag{5}$$

with $\ell_P = \sqrt{\hbar G/c^3}$. The associated horizon temperature is written in the standard form

$$T_h = \frac{\hbar c}{2\pi k_B R_h}. \tag{6}$$

These are imported physical bridge inputs, not derived here.

Boundary-area suppression

The horizon entropy counts area-like degrees of freedom. The ratio between the horizon scale and the Planck scale gives

$$\frac{\rho_{\rm HM}}{\rho_{\rm Pl}} \sim \left(\frac{\ell_P}{R_h}\right)^2. \tag{7}$$

For the present cosmic horizon, $R_h \sim 10^{26}$ m and $\ell_P \sim 10^{-35}$ m, so

$$\left(\frac{\ell_P}{R_h}\right)^2 \sim 10^{-122}. \tag{8}$$

This is the naturalness core of FDS-X1: the smallness is a boundary-area ratio, not an unexplained cancellation of independent volume terms.

HORIZON-MAINTENANCE SCALING

Thermodynamic horizon energy scale

The simplest horizon-thermodynamic estimate multiplies entropy by temperature:

$$E_h \sim T_h S_h. \tag{9}$$

Using Eqs. (5) and (6),

$$E_h \sim \frac{\hbar c}{2\pi k_B R_h} \cdot \frac{k_B(4\pi R_h^2)}{4\ell_P^2} \tag{10}$$

$$\sim \frac{\hbar c R_h}{2\ell_P^2} = \frac{c^4 R_h}{2G}. \tag{11}$$

Distributed over a horizon volume $V_h = 4\pi R_h^3/3$, this gives

$$\rho_h \sim \frac{E_h}{V_h} \sim \frac{3c^4}{8\pi G R_h^2}. \tag{12}$$

If $R_h = c/H$,

$$\rho_h \sim \frac{3c^2H^2}{8\pi G}. \tag{13}$$

This is the critical-density scale. An equivalent derivation clarifies the power-to-density conversion. Horizon-maintenance power scales as $P_H \sim T_h S_h/\tau_h \sim c^5/G$, the canonical Planck power. Over one Hubble time $\tau_h \sim H^{-1}$, the total maintenance energy is $E_{\rm main} \sim P_H/H$. Distributed over the Hubble volume $V_H \sim c^3/H^3$, the energy density becomes

$$\rho_{\rm HM} \sim \frac{P_H/H}{V_H} \sim \frac{c^5/G}{H} \cdot \frac{H^3}{c^3} = \frac{c^2H^2}{G}, \tag{14}$$

recovering the same $M_{\rm Pl}^2 H^2$ scale. This step makes explicit that the density follows from dividing a per-Hubble-time maintenance energy by the causal volume, closing a possible dimensional attack on the raw $T_h S_h/V_h$ estimate. FDS-X1 identifies the dark-energy component as the fraction or response of this horizon-scale budget associated with boundary maintenance.

Maintenance response factor

Define the horizon-maintenance density

$$\rho_{\rm HM}(a) = \mu(a)\,\frac{3c^4}{8\pi G R_h^2(a)}. \tag{15}$$

When $R_h \sim c/H$, this becomes

$$\rho_{\rm HM}(a) = 3M_{\rm Pl,red}^2 H^2(a)\,\mu(a), \tag{16}$$

where $M_{\rm Pl,red}^2 = (8\pi G)^{-1}$ in natural units. The dimensionless function $\mu(a)$ is a horizon-maintenance response factor. It may encode the fraction of the horizon-scale budget appearing as the effective dark-energy sector, the transition from Hubble-like to event-horizon-like behavior, interaction with matter or curvature, or the mapping from physical horizon maintenance to the fitted dark-energy variable.

Remark 1. *Equation (16) is a model family, not a completed microphysical theory. A full model must specify $R_h(a)$ and $\mu(a)$, derive perturbations, and confront CMB, BAO, supernova, weak-lensing, and growth data.*

Why this is not ordinary vacuum-energy counting

Naive vacuum-energy estimates count independent bulk modes up to a UV cutoff. The FDS-X1 scaling counts accessible horizon distinctions. The difference is schematic:

$$\rho_{\rm bulk} \sim \Lambda_{\rm UV}^4, \qquad \rho_{\rm HM} \sim \frac{A_h}{\ell_P^2}\frac{k_B T_h}{R_h^3} \sim \frac{c^4}{GR_h^2}. \tag{17}$$

The second expression is boundary-like. Its smallness follows from the cosmic horizon being large.

EQUATION OF STATE AND PHENOMENOLOGICAL MODEL

Continuity equation

For a separately conserved effective component,

$$\frac{d\rho_{\rm HM}}{d\ln a} + 3(1+w_{\rm HM})\rho_{\rm HM} = 0. \tag{18}$$

Therefore

$$w_{\rm HM}(a) = -1 - \frac{1}{3}\frac{d\ln\rho_{\rm HM}}{d\ln a}. \tag{19}$$

Using Eq. (16), in the $R_h \sim c/H$ phenomenological regime,

$$w_{\rm HM}(a) = -1 - \frac{2}{3}\frac{d\ln H}{d\ln a} - \frac{1}{3}\frac{d\ln\mu}{d\ln a}. \tag{20}$$

This equation shows why the response factor matters. If μ is constant while H decreases, then $w_{\rm HM} > -1$ and the component tracks the dominant background too strongly. To obtain a late-time component near -1, the horizon response must evolve as the relevant boundary approaches an event-horizon or de Sitter-like limit.

Non-phantom physical component

The strict FDS-X1 expectation is not a precise CPL pair (w_0, w_a). It is the sign of physical density evolution:

$$\frac{d\rho_{\rm HM}}{d\ln a} \leq 0 \quad\Longleftrightarrow\quad w_{\rm HM} \geq -1. \tag{21}$$

At late times, if the universe approaches de Sitter-like horizon stabilization, then

$$\frac{d\rho_{\rm HM}}{d\ln a} \to 0, \qquad w_{\rm HM} \to -1. \tag{22}$$

This is the most conservative dynamical claim: the physical horizon-maintenance component should be non-phantom and should approach -1 as the effective horizon stabilizes.

Effective reconstructions versus physical component

Cosmological data often constrain parameters in an effective model, such as the CPL form

$$w(a) = w_0 + w_a(1-a). \tag{23}$$

A fitted $w_{\rm eff} < -1$ in such a model need not by itself be the physical equation of state of the horizon-maintenance sector. It may arise from parameterization, sector mixing, curvature, modified-gravity mapping, data systematics, or the fact that $\mu(a)$ and $R_h(a)$ have not been explicitly modeled. FDS-X1 therefore distinguishes:

1. the *strict physical claim*: the horizon-maintenance component is non-phantom;
2. the *effective reconstruction*: a data-fitting variable that may cross -1 without directly representing a fundamental phantom fluid.

This distinction is not an unlimited escape clause. If nonparametric reconstructions and physical model comparisons robustly require a true phantom component with no sector-mixing or mapping ambiguity, the strict version fails.

PRE-EUCLID PREDICTIONS

Prediction 1 (P1: scale)**.** *The dark-energy sector should remain tied to a horizon-scale density of order $M_{\rm Pl}^2 H_0^2$, or equivalently $c^2H_0^2/G$ in SI units, not to a Planck-volume vacuum-energy density.*

Prediction 2 (P2: non-constant physical sector)**.** *The physical horizon-maintenance sector should not be a featureless bare constant at all epochs. It should encode horizon evolution through $R_h(a)$ and/or $\mu(a)$, even if the late-time limit is observationally close to $w = -1$.*

Prediction 3 (P3: non-phantom physical behavior)**.** *The strict physical horizon-maintenance component should satisfy $w_{\rm phys} \geq -1$, or equivalently $d\rho_{\rm HM}/d\ln a \leq 0$, with approach toward -1 as the effective horizon stabilizes.*

Prediction 4 (P4: effective phantom allowance)**.** *If Euclid, DESI, or combined data prefer $w_{\rm eff} < -1$ in CPL or similar fits, FDS-X1 treats this as compatible only if the phantom behavior is effective rather than fundamental: for example, due to sector mixing, parameterization, curvature, modified-gravity mapping, systematics, or incomplete modeling of $\mu(a)$ and $R_h(a)$.*

Prediction 5 (P5: density reconstruction priority)**.** *The most direct observational test is not the sign of a fitted w_0 alone but the reconstructed evolution of $\rho_{\rm DE}(a)$. A robust increase of physical dark-energy density with scale factor challenges the strict non-phantom horizon-maintenance version.*

FALSIFICATION AND DEMOTION CONDITIONS

No ad hoc rescue rule

FDS-X1 is not allowed to treat every possible observational result as success. If future data strongly support a strict cosmological constant with no density evolution, the dynamical claim is demoted. If future data robustly require a physical phantom component, the strict non-phantom version is falsified. If the horizon-scale density relation remains merely dimensional and cannot be embedded in a perturbatively viable cosmological model, the claim remains a heuristic naturalness argument rather than a physical theory.

Post-Euclid update protocol

After Euclid DR1, the status update should use the same categories as Table III:

1. unchanged frontier bridge;
2. promoted scaling/naturalness bridge;
3. supported dynamical non-phantom version;
4. demoted to heuristic only;
5. strict version falsified;
6. retired.

The update should quote the pre-Euclid predictions before interpreting the data.

COMPARISON WITH EXISTING MODELS

Versus ΛCDM

ΛCDMtreats dark energy as a constant density with $w = -1$. FDS-X1 accepts ΛCDMas the current benchmark and does not claim observational replacement. Its difference is interpretive and predictive: it treats the observed scale as horizon-maintenance scale and expects that sufficiently precise data may reveal controlled evolution in the physical dark-energy sector.

Versus standard holographic dark energy

Standard holographic dark-energy models typically choose an infrared cutoff L and posit

$$\rho_{\mathrm{HDE}} = 3c_{\mathrm{HDE}}^2 M_{\mathrm{Pl,red}}^2 L^{-2}. \tag{24}$$

The physics depends strongly on the choice of L. Taking $L = H^{-1}$ has known difficulties in simple models; taking L as the future event horizon can produce acceleration [18, 19]. FDS-X1 agrees that the event horizon or another causal boundary may be the physically relevant late-time scale, but it gives the cutoff a finite-distinguishability interpretation: the horizon is an operational boundary of accessible distinctions, not merely a regulator.

Versus quintessence and phantom models

Quintessence introduces scalar-field dynamics to obtain $w \neq -1$, usually with $w > -1$. Phantom models permit $w < -1$ but often face theoretical difficulties. FDS-X1 is not a scalar-field model. Its strict physical expectation is non-phantom, while allowing effective phantom reconstructions only as descriptions of a mixed or mis-specified sector.

Versus modified gravity

Modified-gravity models may reproduce effective dark-energy behavior without a physical fluid. FDS-X1 remains compatible with the possibility that some observed w_{eff} behavior is a mapping of modified gravitational dynamics into an effective fluid. A full FDS-X1 model must therefore be tested not only against background expansion but also against perturbations and growth of structure.

LIMITATIONS

First, FDS-X1 is a frontier physical consequence (P3 layer). It is not established by FDS-T1 alone. Failure of X1 does not falsify the FDS formal core or the finite-observer budget paper.

Second, the scaling $\rho_{\mathrm{HM}} \sim M_{\mathrm{Pl}}^2 H^2$ is not by itself a full cosmological model. A constant coefficient with $L = H^{-1}$ has known problems in ordinary holographic dark-energy treatments. The response factor $\mu(a)$ and the appropriate horizon radius $R_h(a)$ must be specified for a complete model.

Third, the paper does not compute perturbations. Any viable model must match CMB anisotropies, BAO, supernova distances, weak lensing, structure growth, and local constraints.

Fourth, the paper does not derive the Bekenstein-Hawking entropy formula or de Sitter temperature. These are bridge inputs.

Fifth, the paper does not claim that the QFT vacuum-energy calculation is mathematically invalid. It claims only that the observed cosmic dark-energy scale may be

TABLE III. Pre-registered outcome table for FDS-X1.

Future result	**Consequence for FDS-X1**
High-precision, model-independent consistency with constant $\rho_{\rm DE}$ and $w=-1$ across the measured redshift range	Demotes the dynamical horizon-maintenance claim; only the naturalness scaling may remain as a heuristic
Mild dynamical dark-energy evidence with $w_{\rm phys} \geq -1$ and late-time approach to -1	Supports the strict horizon-maintenance version, subject to model comparison and perturbation tests
CPL or other effective fits prefer phantom crossing, but nonparametric density reconstructions or physical models do not require a fundamental phantom component	Compatible with the effective-reconstruction allowance; requires explicit modeling of $\mu(a)$, $R_h(a)$, curvature, and sector mixing
Robust model-independent evidence that physical $\rho_{\rm DE}$ increases with scale factor	Strongly challenges or falsifies the strict non-phantom version
Robust unavoidable physical $w < -1$ with no mapping, mixing, modified-gravity, or systematic ambiguity	Falsifies the strict non-phantom horizon-maintenance component
Data remain inconclusive or strongly model dependent	FDS-X1 remains a frontier physical consequence; no promotion

associated with accessible horizon-boundary degrees of freedom rather than independent bulk modes.

Sixth, the non-phantom expectation applies to the physical horizon-maintenance component. Effective fitted parameters may differ if the model space is incomplete.

CONCLUSION

FDS-X1 registers a pre-Euclid frontier physical consequence: if cosmological horizons are finite distinguishability boundaries, then their maintenance naturally carries the density scale

$$\rho_{\rm HM} \sim \frac{c^4}{G R_h^2} \sim M_{\rm Pl}^2 H^2, \tag{25}$$

with smallness

$$\frac{\rho_{\rm HM}}{\rho_{\rm Pl}} \sim \left(\frac{\ell_P}{R_h}\right)^2 \sim 10^{-122}. \tag{26}$$

This reframes the dark-energy naturalness problem as a boundary-area suppression rather than a cancellation of bulk vacuum terms.

The note does not claim to solve cosmology. It registers a model family and an outcome table before Euclid DR1. The strict physical prediction is a dynamical but non-phantom horizon-maintenance component with late-time approach toward -1. Effective phantom fits are allowed only if they are reconstructions of a mixed or misspecified sector. Robust constant dark energy demotes the dynamical claim; robust physical phantom behavior falsifies the strict version.

Future work must specify $R_h(a)$, derive $\mu(a)$, compute perturbations, fit DESI, Euclid, Roman, supernova, CMB, and weak-lensing data, and compare against ΛCDM, holographic dark energy, quintessence, phantom models, interacting dark energy, and modified gravity.

The author used AI-assisted tools for language polishing, structural feedback, and drafting support. All claims, arguments, citations, and final wording remain the responsibility of the author.

Notation Summary

Claim Status and Failure Conditions

Boundary of Applicability

The FDS-X1 bridge applies only under the following conditions:

1. a meaningful cosmological horizon or finite causal-access boundary exists;
2. the horizon entropy and temperature bridge inputs are valid in the relevant regime;
3. the observed dark-energy sector can be compared to an effective horizon-maintenance density;
4. the mapping from physical $\rho_{\rm HM}$ to fitted $w(a)$ is not assumed to be trivial unless specified.

The bridge does not apply as stated in regimes where no finite horizon boundary exists, where horizon thermodynamics fails, or where the dark-energy phenomenology is entirely due to a sector unrelated to causal-boundary maintenance.

Priority and Timestamp Notice

This document is intended as a pre-Euclid public registration of the FDS-X1 frontier physical consequence.

TABLE IV. FDS-X1 notation summary.

Symbol	Meaning
R_h	effective cosmological horizon radius used by the bridge model
$R_H = c/H$	Hubble radius
R_e	future event horizon radius
$A_h = 4\pi R_h^2$	horizon area
S_h	horizon entropy
T_h	horizon temperature
ρ_{HM}	horizon-maintenance energy density
ρ_{DE}	observationally reconstructed dark-energy density
$\mu(a)$	horizon-maintenance response factor
w_{HM}	physical equation of state of the horizon-maintenance component
w_{eff}	effective fitted equation of state in a chosen reconstruction
ℓ_P	Planck length
$M_{\mathrm{Pl,red}}$	reduced Planck mass
Ω_{DE}	dark-energy density fraction

TABLE V. Claim status and failure conditions for FDS-X1.

Claim	Status	Failure condition
Horizon distinguishability boundary	Physical bridge	Horizon entropy or causal-boundary accounting fails in the applicable regime
$\rho_{\mathrm{HM}} \sim M_{\mathrm{Pl}}^2 H^2$	Scaling bridge	Dark-energy scale shown unrelated to horizon or causal-boundary scale
Area-ratio suppression	Naturalness bridge	Observed scale requires an unrelated cancellation mechanism
Non-phantom physical component	Pre-Euclid prediction	Robust unavoidable physical $w < -1$
Late-time -1 approach	Asymptotic bridge	Persistent future departure incompatible with horizon stabilization
Effective phantom allowance	Interpretation rule	Effective reconstructions proven to represent a true phantom sector

Its purpose is not to assert that the claim is established, but to record the statement, dependencies, predictions, and failure conditions before Euclid DR1. Later versions may revise, demote, or abandon claims according to the outcome table.

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